SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I M.VISHAL. (192565077) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
M.Vishal.

19/20

Annexure A

SCENARIO BASED QUESTIONS:

Smart Campus Subnet Allocation:

Identification of Key Issues.

* Allocate Subnets Using the Given Subnet masks
* Find the Network and Broadcast addresses
* Identify the valid host IP address Range
* Calculate the Number of unsable host addresses
* Determine the remaining unsend IP address Range.

Applications of Concepts:

Department	Subnet Mask	Network Address	Valid Host Range
School of Computing	/25	192.168.100.0	192.168.100.1-126
School of Electronics	/26	192.168.100.128	192.168.128.129-190
School of Mechanical	/27	192.168.100.136.	192.168.100.137-222

Remaining unused IP Address Range:

* 192.168.100.224-255
* Total remaining address : 32

Decision making

The Subnet allocation efficiently Satisfies the Requirement of all three department while avoiding IP address overlap. The remaining address block is reserved for future expansion.

Conclusion;

The Smart campus Network has been Successfully divided into three Subnets based on the required Subnet masks

2. Multinational Software Company Subnet Allocation:

Identification of Key Issues:

* Allocate Subnets using the given Subnets marks
* Determine network and broadcast addresses
* Find the valid host IP ranges
* Calculate the unsend IP address Range.

Application of Concepts:

Department	Subnet	Network Address	Host Range	usable Host
Software Development	/24	10.10.0.0	10.10.0.1 - 254	254
Quality Assurance	/25	10.10.1.0	10.10.1.1 - 126	126
Human Resources	/26	10.10.1.128	10.10.1.129 - 190	62
Executive Managment	/27	10.10.1.192	10.10.1.193 - 222	30

Decision Making:

The Subnet allocation provides sufficient IP address for Each department and allows future expansion with the Remaining unused address Space.

Conclusion:

The Company network is efficiently divided into four Subnets with proper host allocation and a large unsend address block for future requirements

* ...ion of key issues
* Compress the IPv6 address
* Expand the compressed address
* Identify the /64 network prefix
Calculate the Total Number of Host addresses

Application of Concepts

* Compressed IPv6 Addresses: 2001:db8:abcd::/64
* Network Prefix: 2001:db8:abcd::/64
* Total Host Addresses 2^{64} =
18,446,744,073,709,551,616.

Decision Making:

A /64 Prefix is the standard IPv6 subnet size and Supports an extremely large Number of Devices, making it suitable for hospital with IOT and Medical equipment for Hospital with IOT and Medical equipment

Conclusion:

The IPv6 Network is Correctly Configured with a /64 Prefix providing a massive address space for Current and future Network expansion.

Routing Table Analysis:

Identifications of Key Issues.

* Identify the destination Subnet
* Match the routin table Entry
* Determine the forwarding interface
* Identify the default route if no match exists

Application of Concepts.

* Destination IP: 192.168.2.45
* Destination Subnet : 192.168.2.0/24
* Matching Routing Entry : 192.168.2.0/24
* Default Route : 0.0.0.0/0

Decision Making:

Since the destination IP belongs to the 192.168.2.0/24 Subnet, the router forwards the packet thoought the interface Connected to that Subnet.

Conclusion:

The packet is Successfully forwarded using the matching routing table Entry.

5 Layer-3 Router LAN Configuration.

Identification of key issues.

* Find the Network and Broadcast address
* Determine valid host Ranges
* Assign IP addresses to computers
* Identify the Correct default gateway

Suggested IP Address.

* PC1 - 192.168.10.2, 192.168.10.3, 192.168.10.4

Decision Making:

The Selected IP Addresses fall within the valid host range and the configured gateways enable Communication between both Lans

Conclusion:

The Administration and Finance LANS are Correctly Configured with Valid Network address, host Range, gateway addresses, and Computer IP Assignments, Ensuring reliable Network Communication.